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Unlocking Production and Revenue with Single Phase Retarded Acid - Results from the First Deployment Phase in an Onshore Field of Middle East

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Abstract

This paper presents the results of the initial deployment phase of a novel stimulation fluid technology known as Single Phase Retarded Acid (SPRA). The objective was to assess its effectiveness in enhancing well performance across a range of well types—oil producers, gas producers, and water injectors—completed in moderate permeability reservoirs. The study aimed to compare SPRA's performance against conventional acid systems in terms of productivity/injectivity gains, treatment longevity, and overall economic impact.

The deployment was carried out in multiple fields, encompassing a representative cross-section of well types of predominantly moderate permeability. A critical part of the process involved extensive laboratory testing, primarily Pore Volume to breakthrough (PVbt), to screen the most suitable SPRA formulations available in the market. These tests provided insight into the acid's wormhole propagation characteristics and reaction retardation behavior. Treatments were designed based on well-specific parameters adhering to consistent pumping schedules ensuring comparability. Pre- and post-stimulation data were gathered through productivity & injectivity monitoring, pressure transient analysis, and well surveillance data. A comparative economic analysis was also conducted to quantify incremental value generation.

Field results demonstrated consistent and notable improvements in well performance following SPRA treatments. Oil and gas producers exhibited significant increases in productivity index (PI), with production enhancements sustained for longer durations than comparable wells treated with conventional acid systems. Water injection wells showed marked improvements in injectivity, accompanied by greater stability and reduced fall-off in performance. Diagnostic data indicated deeper stimulation penetration and improved acid placement, contributing to more effective and efficient wormholes propagation. The economic analysis revealed substantial revenue uplift and reduced the need for re-stimulation, underscoring the technology's potential for improving asset value. Additional value was identified from the technology in other aspects of the stimulation, like potential self-diversion and dosage reduction.

This is the first documented field-scale deployment of SPRA technology across multiple well types. The combination of lab-based PVbt screening and field implementation offers a novel, data-driven approach to acid system selection and treatment design. SPRA demonstrated flexibility across various reservoir conditions while delivering superior and sustained well performance. These results position the technology as a promising alternative for wider application in moderate permeability formations.

Introduction

Acid stimulation remains a key method to enhance well productivity and injectivity in carbonate reservoirs, however in moderate permeability formations of high temperature, the use of conventional hydrochloric acid (HCl) often leads to rapid acid-rock reaction rates, resulting in inefficient wormhole development, shallow penetration, and limited treatment durability (Gdanski, 1997; Nasr-El-Din et al., 2002).

To address these challenges, a variety of acid retardation strategies have been developed over the years. These include gelled acids (e.g., crosslinked or polymer-thickened systems), emulsified acids, microemulsions, and encapsulated acid systems, each designed to slow reactivity and improve acid placement in the formation or "wormhole" development (Bartko and Wills, 1991; Kalfayan, 2008). While such systems have shown improvements in certain applications, they can introduce operational complexity, formation damage risk, or post-treatment residue cleanup issues.

In response, Single Phase Retarded Acids (SPRA) emerged in the early 2000s as a simpler and cleaner alternative. These formulations use retardation additives to control the diffusion of reactive ions, thereby reducing the acid's effective reaction rate without requiring phase separation or gelling (Nasr-El-Din et al., 2003; Al-Mutairi et al., 2010). SPRA systems maintain a single aqueous phase, enabling deeper acid penetration, uniform stimulation, and minimal residue, all while being compatible with standard pumping and operational procedures.

More recently, Morrow et al. (2024, 2025) conducted extensive laboratory research on commercial SPRA formulations using Pore Volume to breakthrough (PVbt) testing in Indiana limestone and dolomite. Their findings established clear performance differentiation among SPRA systems, highlighting that optimized SPRA formulations significantly outperform unretarded HCl in wormhole efficiency and depth of stimulation. Moreover, the second phase of their study demonstrated that lower retarder concentrations could achieve comparable PVbt performance, pointing to cost-effective design opportunities in field applications.

Building on this laboratory foundation, this paper presents the first multi-well deployment of SPRA technology across oil and gas producers and water injectors in moderate permeability carbonate reservoirs. The evaluation incorporates PVbt testing for acid selection, well-specific treatment design, and post-stimulation surveillance and diagnostics, as well as a comparative assessment against conventional acid systems. Results demonstrated consistent improvements in well performance, extended treatment longevity, and meaningful economic uplift, supporting SPRA as a viable and high-value stimulation solution in carbonate reservoir development.

Methodology

The methodology adopted in this study was designed to systematically evaluate the performance of Single-Phase Retarded Acids (SPRA) under field conditions and benchmark it against conventional acid systems. The workflow included Laboratory Testing and Technology Selection, Well Selection and Data Gathering and Performance Evaluation Framework.

Laboratory Testing

A comprehensive summary of top-performing SPRA systems for both calcite and dolomite reservoirs, identified through extensive PVbt laboratory testing by Morrow et al. (2024, 2025a) and collaborators,

forms the scientific basis for SPRA selection and recommendations in this study; the detailed findings and methodology behind these selections will be discussed in the following section.

Well Selection and Data Gathering

Candidate wells were selected based on a combination of geological and operational parameters. These included:

- Reservoir permeability (between 0.5 and 10 md)
- Well with severe declining performance
- Wells with unsuccessful results with unretarded HCl systems
- High Reservoir Temperature
- Availability of pre-stimulation production or injectivity data
- Well integrity and operational readiness
- Diversity in well types (oil, gas, and water injectors)

Each candidate underwent a pre-job surveillance phase that included pressure transient analysis, production history review, and previous stimulation records (if available).

Performance Evaluation Framework

To evaluate the effectiveness of SPRA treatments, a pre-defined monitoring and evaluation protocol was implemented. This included:

- Pre- and post-stimulation injectivity/productivity index (PI) comparison
- Normalized production/injection rate trends
- Pressure transient testing (when feasible)
- Well test and surveillance data integration
- Benchmarking against historical acid treatments in offset wells

Economic metrics such as incremental revenue, avoided re-stimulation cost, and NPV uplift were also calculated to assess the commercial impact.

Laboratory Testing and Technology Selection

The work of [Morrow et al. \(2024, 2025a\)](#) reported a list of top-of-class SPRA systems for acidizing calcite reservoirs based on PVbt data in Indiana Limestone cores, and a separate list of top-of-class SPRA systems for acidizing dolomite reservoirs based on PVbt data in Bonne Terre Dolomite cores. These lists of top-of-class SPRAs provide the basis for most of the recommendations for SPRA systems to be trialed in this work. In some cases, PVbt tests with 2 or 3 top-of-class SPRAs were carried out on reservoir cores before making a final recommendation on which specific SPRA to trial in a specific reservoir ([Al Yaqoubi et al., 2024](#); [Alkhaldi et al., 2025](#); and [Morrow et al., 2025b](#)). In this work, we use the identical SPRA nomenclature to that of [Morrow et al. \(2025b\)](#):

1. System A04: The identical 15% HCl based system tested in SPE-222481
2. System B04: The identical 20% HCl based system tested in SPE-225375, labeled there as A04 (20% HCl).
3. System A05: The identical 15% HCl based system tested in SPE-222481.

4. System B05: The identical 28% HCl based system tested in SPE-225375, labeled there as A05* (28% HCl).
5. System A06: The identical 15% HCl based system tested in SPE-222481.
6. System A07: The identical 15% HCl based system tested in SPE-222481
7. System A08: The identical 15% HCl based system tested in SPE-222481.

Detailed descriptions of the laboratory PVbt testing of the SPRA systems listed above in quarry cores can be found in SPE-222481 and SPE-225375. In summary, 1.5"-diameter, 6"-length Indiana Limestone and Bonne Terre dolomite cores were used to perform comparative testing of the performance of commercial SPRA systems. 9 different 15% HCl-based SPRAs were tested on Indiana Limestone cores and 8 different SPRAs, with base HCl concentration of either 15% HCl, 20% HCl, or 28% HCl were tested on Bonne Terre Dolomite cores. Complete acid recipes were not used for these experiments, instead, only the special retarder additives were added to the base HCl. Each SPRA system was tested at multiple injection rates to construct a complete PVbt curve. The SPRAs with the lowest PVbt values relative to unretarded HCl were deemed to be the best performing products. All experiments on Indiana Limestone cores used a test temperature of 250°F and unretarded 15% HCl as the baseline system for comparison. All experiments on Bonne Terre Dolomite cores used a test temperature of 270°F and unretarded 28% HCl as the baseline system. A back pressure of 3000 psi and net confining pressure of ~800 psi were used in all the experiments.

The Indiana Limestone and Bonne Terre Dolomite testing did not show any one SPRA as the absolute best product. Instead, multiple SPRAs that showed similar PVbt performance relative to the baseline unretarded acid were reported as the "best-in-class". For the study using Indiana Limestone, SPRAs A04, A06, and A07 were reported to be among the best-in-class systems for limestone reservoirs. The best-in-class SPRAs for limestone reservoirs were predicted to give 2-5X better stimulation performance over unretarded 15% HCl in (high-rate) bullheading treatments, and as much as 10X better performance for (low-rate) coiled tubing stimulation treatments. SPRA A05 was noted as showing notably better performance than unretarded 15% HCl in quarry core testing despite not being among the best-in-class. SPRA system A08 showed some improved performance relative to unretarded 15% HCl but was not labelled as a best-in-class SPRA for limestone reservoirs. However, the acid recipe ingredients for system A08 are produced in-country and are provided through a local supplier. Therefore, there is value in evaluating the performance of system A08 through field trials. For the Bonne Terre Dolomite study, systems B04, B05, and B06 were reported to be part of the best-in-class for dolomite reservoirs. The best-in-class SPRAs for dolomite reservoirs were predicted to provide 1.6-5X better stimulation performance over unretarded 28% HCl in bullheading treatments, and as much as 9.5X better performance in coiled tubing treatments.

In addition to these benchmarking studies in quarry core plugs, some testing of SPRA systems on core plugs from onshore Abu Dhabi reservoir plugs has been conducted. Al Yaqoubi et al. (2004) reported PVbt testing of two 15% HCl-based SPRAs in three different reservoir rock types. This study showed that in one reservoir rock type, the SPRAs did not appear to provide significant improvement over unretarded 15% HCl, while the SPRAs did appear to provide better stimulation performance in the other two rock types. Due to the relatively small number of PVbt experiments in this study, no conclusion was made regarding which SPRA system would provide the best performance. Alkhaldi, et al. (2025) reported PVbt testing of two chelating agent stimulation systems alongside one 20% HCl-based SPRA and one 28% HCl-based SPRA on a small number of quarry cores and reservoir cores. This study showed that the chelating agent-based systems could produce good wormhole structures in tight cores, but at the cost of significant injection volume. Only 4 experiments were performed with the SPRA systems, but the PVbt results were good enough to drive the decision to trial an SPRA system in the high-temperature, tight oil reservoir described in that paper. More recently Morrow et al. (2025b) reported the results of 74 PVbt experiments on various Abu Dhabi limestone and dolomite reservoir cores involving 10 SPRA systems. This study noted that the superior performance of

the best-in-class SPRAs over unretarded acid systems that was observed in SPE-222481 and SPE-225375 did not carry over to all of the reservoir rock types studied. The authors reported the following observations:

- To properly evaluate the benefits of SPRAs, reservoir cores must be used in laboratory PVbt testing.
- SPRAs are justified for most reservoirs when low pumping rate is expected (e.g. coiled tubing treatments).
- SPRAs do not appear justifiable when high pumping rate is expected (e.g. bullheading treatments).

Treatment Design

SPRA stimulation treatments were designed using a structured engineering workflow aimed at standardization and comparability across different wells while maintaining customization to local conditions. The main elements included:

Treatment Volume and Pumping Schedule

As bottom line the treatment dosages and volumes design started from the guidelines used for conventional unretarded systems (Table 1); additional work was performed with Matrix Acidizing simulators, incorporating the data obtained from the PVbt testing (Figure 1), and optimizing wormhole length and skin evolution (Figure 2).

Table 1—Design Starting Acid Dosage

Type of Treatment	Minimum Acid Dosage (gal/ft)	Maximum Acid Dosage (gal/ft)
Coiled Tubing - Cased hole Perforated (vertical or deviated wellbore)	75	420
Bull head - Cased hole Perforated (vertical or deviated wellbore)	25	420
Coiled Tubing - Open Hole	10	20
Coiled-Tubing - Pre-perforated Liner	10	25
Bull head - Open hole	25	42
Bull head - SMART LINER/ICD/ICV	20	42

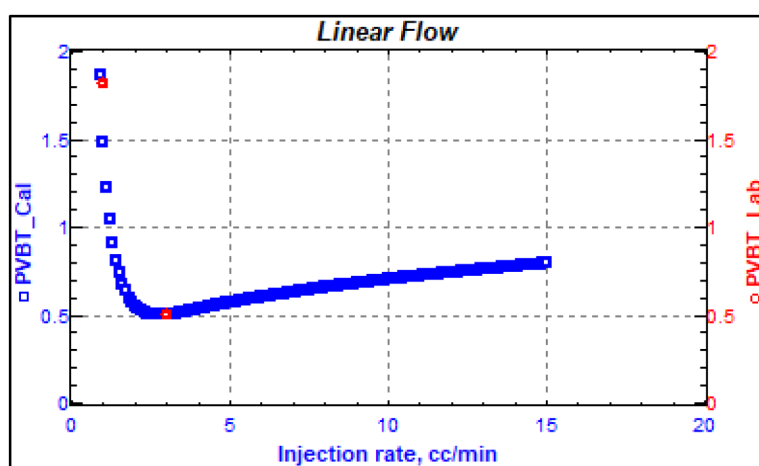


Figure 1—Matrix Acid Simulator - PVbt Data Calibration (Red Points = Lab Data)

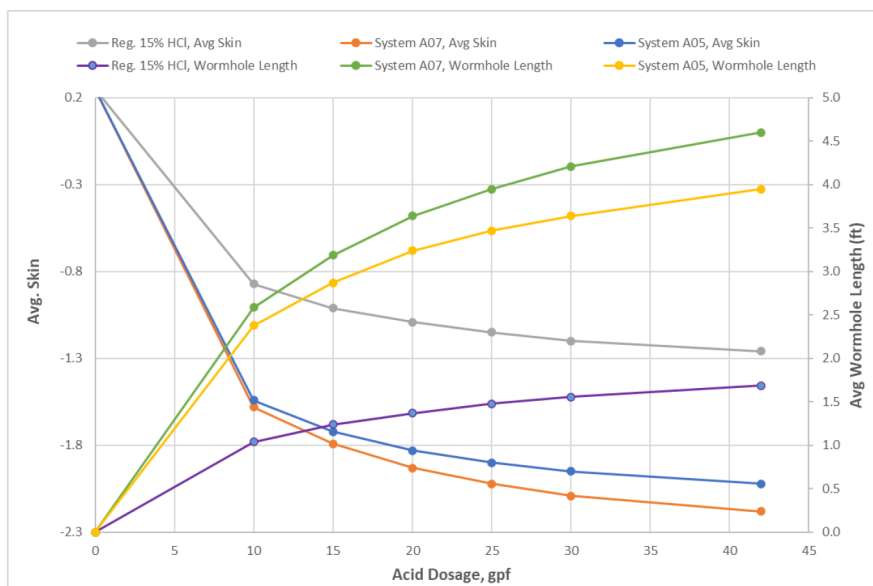


Figure 2—Acid Dosage Optimization Example

For the pump schedules, a three-stage approach was typically followed:

1. Preflush – to displace brine and remove near-wellbore solids. In some occasions the pre-flush involves additional stages of hydrocarbon solvents, aimed to remove organic material in the wellbore and the formation, preparing also the last one to better take the inorganic stimulation
2. Main SPRA stage – injected at matrix rates (typically 1–3 bpm), ensuring controlled contact and deep propagation
3. Overflush – to displace the acid from the tubing and help push the reacted acid deeper into the formation

The absence of emulsions or gelling agents allowed the use of standard surface equipment and simplified logistics.

Additive Compatibility and Fluid Quality Check

In addition to the core PVbt benchmarking, further laboratory testing was conducted on the final acid recipes for field application, incorporating all additives required for operational deployment. This protocol mirrored the standard qualification process employed for conventional stimulation jobs utilizing straight HCl. Specifically, comprehensive evaluations of corrosion and compatibility were performed, with the latter encompassing both emulsion and sludge tendency testing. These compatibility tests were restricted to oil producer wells, reflecting their heightened susceptibility to emulsion and sludge issues. To qualify for field use, each system was required to meet strict acceptance criteria: corrosion rates had to remain below 0.05 lb/ft², emulsion testing demanded less than 3% emulsion after 30 minutes, and sludge formation was limited to a maximum of 1 gram per 50 milliliters of oil. These benchmarks ensured that the selected acid systems would provide both effective stimulation and operational reliability in the targeted reservoir environments.

Job Execution and QA/QC

Jobs were executed using standard pumping units and monitored in real-time. Key quality control steps included:

- Verification of fluid properties and concentrations on location
- Use of calibrated surface equipment for pressure and rate control

- Sample collection for post-job acid residual analysis
- Data logging for post-treatment evaluation and modeling

Operational learning was continuously integrated into subsequent treatments, leading to progressive optimization of the treatment parameters across the deployment phase.

Case Histories

Case 1: Oil Producer (OP-01) and Water Injector (WI-01) in Thin Layers Low Permeability Reservoir

The below case studies from one of the fields in our company were already published by Maheshwari et al. (2024)., where A07 Retarded system was used to successfully reactivate two wells, an oil producer (OP-01) and a Water Injector (WI-01), both in one of the most challenging onshore tight carbonate reservoirs.

The oil producer over time experienced a production decline of 50%, whereas the water injector was injecting almost zero, indicating severe formation damage in both the wells. The SPRA stimulation was combined with fiber optic-equipped coiled tubing for distributed temperature surveying to treat damaged horizontal oil and water wells.

An advanced stimulation approach was applied using the Matrix Stimulation Engineering (MSE) workflow to help ensure uniform acid distribution across the entire lateral (Figure 3). This technique allowed better acid placement, deeper acid penetration and more effective wormhole formation leading to significant improvements in both hydrocarbon production and water injection performance.

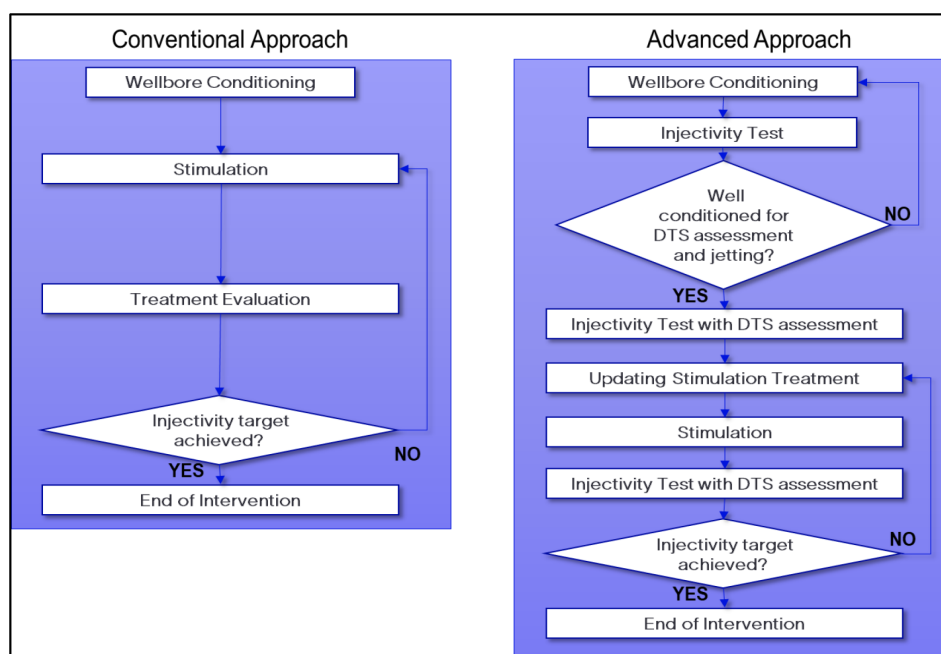


Figure 3—Comparison between the conventional and the advanced stimulation approaches (Maheshwari et al. (2024))

After the intervention, oil production almost doubled for well OP-01, whereas the water injector WI-01 was reactivated with a 14-fold increase (FOI) in the Injectivity Index. Attempts with conventional stimulation resulted in minimum incremental production and injection for these wells. The financial indicators for these wells are summarized in Table 2. From them the incremental cost and revenue correspond to the differential values with respect to the unretarded HCl.

Table 2—Treatment Financial Indicators for Wells OP-01 & WI-01

Well	Incremental Prod with SPRA	Incremental Revenue year 1	Incremental Cost	5 year –NPV Increment
OP-01	1.8 FOI	17%	2.5 X	16%
WI-01	14 FOI	15%	1.5 X	13%

Case 2: Oil Producer in Moderate Permeability Reservoir

The OP-02 well, drilled in an onshore limestone reservoir of moderate permeability, offers another compelling example of targeted engineered SPRA stimulation, leading to significant performance improvements. This reservoir is characterized by a average porosity of 18% and a average permeability of 1.2 mD, with an original reservoir pressure of 4,425 psi and a temperature of 255°F. The OP-02 is a horizontally drilled oil producer, featuring a 3,100-foot lateral completed with a 6-inch open hole.

Upon initial completion, the well demonstrated promising production with a moderate water cut of 13%. Over time, a marked decline ensued, with production rates falling to a fifth of its initial rate and the well frequently becoming inactive and only intermittently producing. The abnormal behavior of this candidate, as compared to neighbors, highlighted the need for a stimulation beyond the normal practices.

The intervention strategy centered on the application of the A05 retarded acid system, utilizing a dosage of 20 gallons of acid solution per foot of horizontal, as per simulation/optimization work. Along with the design process, some risky zones in the horizontal were identified with some potential for water production after stimulation. Adjustments to treatment involved the targeting of only the upper half of the well, as well as the incorporation of chemical diversion stage before the main acid, to reduce the chance to stimulate these unwanted interval(s). Deployment was executed using an abrasive jetting tool conveyed on coiled tubing, pursuing a more precise acid placement along the lateral.

The treatment was executed as planned with a total volume for the main acid of 785 bbl, placed at a maximum rate of 1.25 barrels per minute. The Pre and Post-Stimulation SRT/Injectivity tests, were early indirect indicators of acceptable stimulation effect (Figure 4).

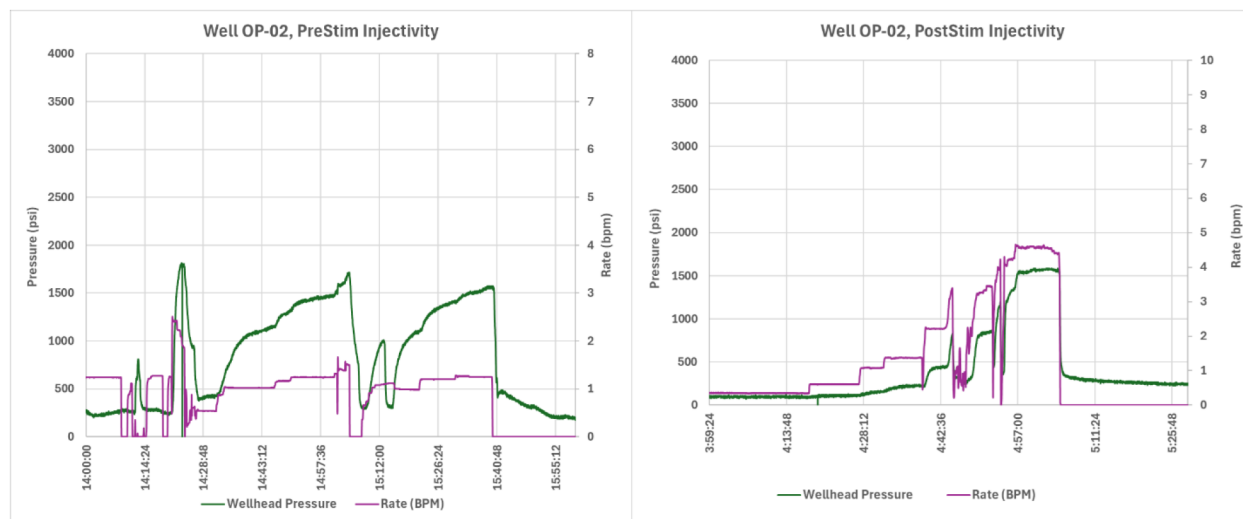


Figure 4—Well OP-01, Pre vs. Post-Injectivity Test

Following the stimulation treatment, the results were also notable, with production rates increased by a factor of 4 FOI as compared to the average pre-stimulation conditions (Comparison at the same wellhead pressure). Figure 5 shows the contrast between an intermittent - abrupt declined Pre-Stimulation production,

with a significantly higher and Stable one after the SPRA treatment. Although additional diagnostic data such as pressure buildup (PBU) or production logging (PLT) were not available, the observed changes in production rates and wellhead pressure provide clear evidence of successful remediation and improved reservoir connectivity.



Figure 5—Well OP-02 Production Performance

The economic analysis of well OP-02 demonstrates the significant value of advanced stimulation using the Single Phase Retarded Acid (SPRA) system. Compared to conventional hydrochloric acid (HCl) treatments—where typical results in the area show a 2-fold increase in production—the SPRA intervention achieved a 4-fold increase, clearly outperforming standard acid jobs in similar reservoir conditions.

Financially, the SPRA treatment resulted in an estimated incremental revenue of 33% in the first year for OP-02, despite the SPRA A05 system costing 2.5 times more than standard HCl. The return on investment (ROI) for the first year grew several times, and the projected incremental five-year net present value (NPV) was 33%. These results illustrate that while SPRA involves higher upfront costs, the operational and economic benefits are substantial, justifying its use.

Case 3: Water Injector in Moderate Permeability Reservoir

Well WI-02 is situated within a moderate-permeability reservoir, traversing clean, laminated limestone characterized by a vertical-to-horizontal permeability ratio (kv/kh) of 0.7. The formation demonstrates favorable reservoir properties, with a porosity of 24%, permeability ranging from 6 to 10 millidarcies, and a reservoir pressure of 2,100 psi. The well was drilled horizontally, extending 2,400 feet with a 6-inch borehole diameter and intersecting multiple stratigraphic layers in an uncentered fashion.

The well was selected for stimulation due to its combination of high wellhead injection pressure (WHIP) and persistently low injectivity, despite prior applications of conventional acid stimulation methods, which offered only minimal and short-lived benefits.

The stimulation program was designed to address these challenges with a carefully engineered sequence. The pre-operational fluid testing involved only corrosion rates confirming the system's robustness, with a measured corrosion rate of 0.0097 lb/ft—substantially below the operational threshold.

The operation was performed in 3 stages:

1. **Injectivity & Sludge Removal:** Pre-Stimulation injectivity test was conducted by the bull-heading technique, at several rates (Step Rate Test). This test was conducted to confirm that enough injection rate is achievable to proceed with the job, as well as establishing the baseline injectivity to be compared with the post job results. The well was also prone to organic deposits buildup in the wellbore, mainly composed of sludge, making it necessary to include an initial stage of 100 bbl of Organic Solvent Stage, which was placed by the bullheading technique, and allowed to soak for 24 hours.
2. **Pre-Flush & Main Acid Treatment:** Subsequently a 5% Mutual solvent solution stage of 55 barrels was bullheaded; CT was then RIH while squeezing additional 95 bbl of solvent solution. The main purpose

of the mutual solvent solution is to displace the solvent, preventing additional sludge formation and retaining formation wettability. SPRA treatment of 85 bbl was applied afterwards from the TD to the liner shoe, and soaked for 1h post over displacement. The CT was again RIH to perform the main acid treatment in the open-hole section from TD-liner shoe, which involved a total volume of 570 bbl of 15% HCl SPRA (10 gal/ft dosage) placed in stations of 10 bbl, every 50 ft interval. The composition of the main acid involved an acid retarder (A08) at a concentration of 5 gallons per thousand gallons.

- Once completed the treatment was over-displaced with fresh water and Post Stimulation Injectivity test was conducted, showing an initial injection rates 1.67X the pre Stimulation rates. Table 3 shows the contrast between the Step Rate/Injectivity Tests conducted during the operation (Prior and Post Pumping) and Figure 6 illustrates the operational sequence previously described.

Table 3—Well WI-02 - Pre-vs Post Injectivity Test

Pre-Stimulation WHCIP = 1200 psi	
Pump Rate	Pressure
1.3 bpm	1200 psi
1.6 bpm	1300 psi
1.8 bpm	1500 psi
Post Stimulation WHCIP = 1300 psi	
Pump Rate	Pressure
2.2 bpm	1300 psi
3 bpm	1500 psi

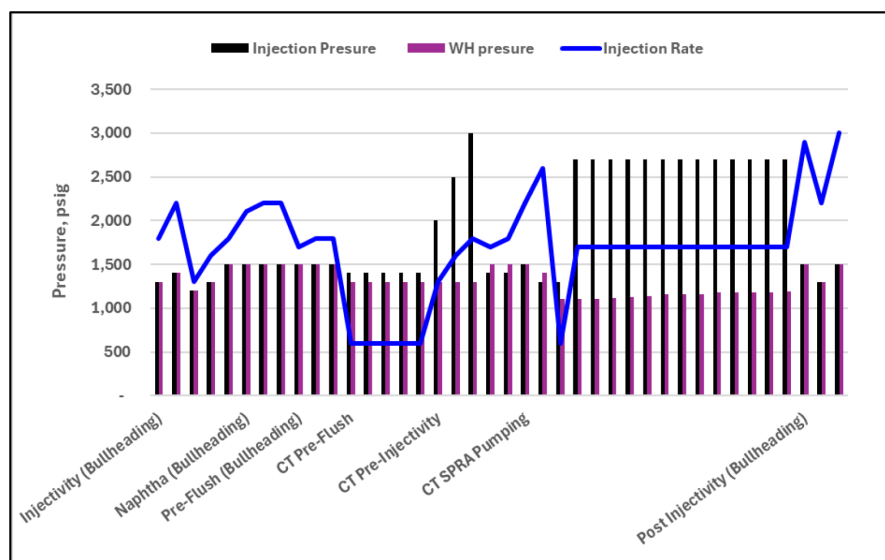


Figure 6—Well WI-02 SPRA Stimulation Operational Sequence

Post well re-commissioning analysis, showed an injectivity improvement of >2.5X. Additional short-term surveillance included Pressure Fall-Off testing, which indicated that the SPRA treatment effectively

improved the pressure draw-down, reservoir connectivity (injection path & new zones), and reduced the flow restriction ([Figure 7](#): Well WI-02 - Pre vs Post SPRA Stimulation Fall Off Test).

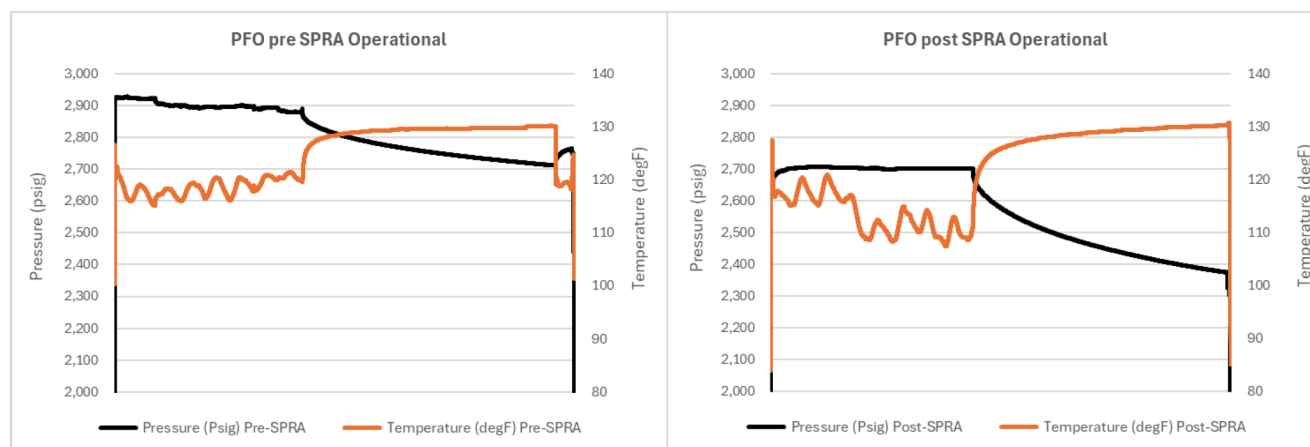


Figure 7—Well WI-02 - Pre vs Post SPRA Stimulation Fall Off Test

The long term production analysis showed a significant contrast, with Pre-stimulation average injectivity of 1X at 1,450 psi, and an Injectivity index of 2.2 STBD/psi, with Post-stimulation average injectivity increased to 2X with a reduced injection pressure of 1,180 psi, and an Injectivity index improved to 3.1 STBD/psi, ([Figure 8](#))

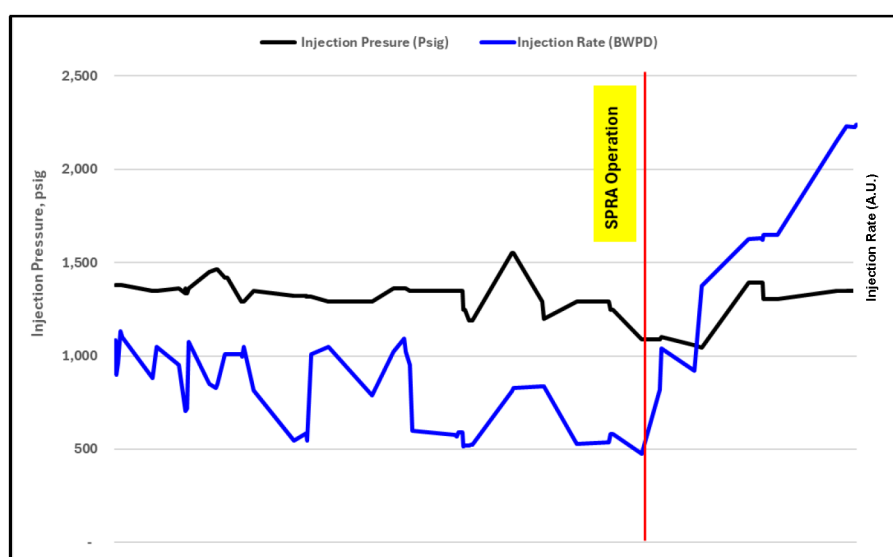


Figure 8—Well WI-02 - Comparative Production Performance Pre and post SPRA Stimulation

In addition, a comparative analysis was conducted with earlier stimulation treatments, each utilizing conventional hydrochloric acid formulations at varying concentrations. The outcomes of these previous interventions are detailed in [Table 4](#) and further illustrated in [Figure 9](#). This comparison clearly highlights the superior and more sustained performance achieved with the implementation of the SPRA treatment, setting it apart from the short-lived or modest improvements observed in prior stimulation events.

Table 4—Well WI-02, Production performance comparison for SPRA and Previous Stimulations

Stimulation Event	Observed Results
15% HCL Stimulation #1	Sharp initial rise followed by a gradual decline
15% HCL Stimulation #2	Short-lived improvement with fluctuations, then trending downward after 6 months
28% HCL Stimulation	Modest rise with no clear sustained trend
Single Phase Retarded 15% HCL (SPRA)	Noticeable and lasting improvement in injection rate

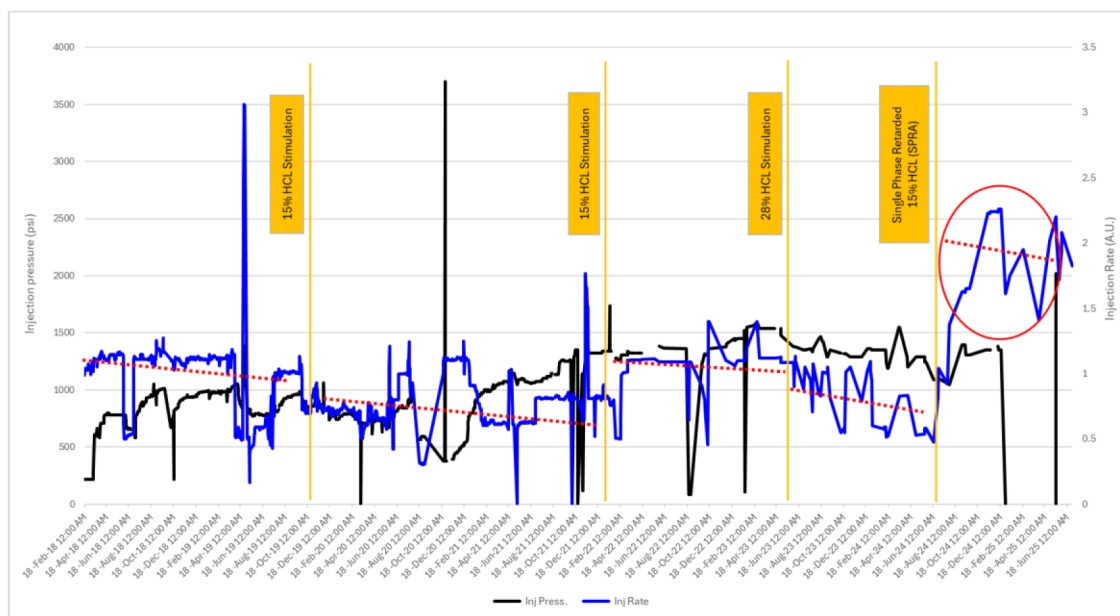


Figure 9—Well WI-02 - Production Performance - SPRA vs Previous Regular Stimulation

For the economic evaluation, the observed injection gain following the SPRA (Single Phase Retarded Acid) treatment was established at 2.0 Folds of Increase (FOI), in contrast to a 1.3 FOI typically achieved with conventional acid stimulation methods. Incremental revenue projections were developed based on the anticipated uplift in oil production observed in near oil producer wells, directly attributed to the additional injected volume. As a result, well WI-01 generated an increase of revenue of ~30% in the first year was calculated.

An important aspect to consider in this case is the incremental cost associated with implementing the A08 retarded acid system, which was only slightly higher than that of unretarded acid systems. Given the relatively low capital investment and notable improvements in both injection rates and production performance, the projected Net Present Value (NPV) grew in the same proportion as the revenue (~30%), positioning this intervention as a highly competitive option for injector well applications in similar reservoir settings. Economic projections and indicators should be re-evaluated in the future using long-term production data to confirm decline rates.

Case 4: Gas Producer Well in Low Permeability Reservoir

Well GP-01 is a vertical, gas-producer completed in the TR-7 reservoir, part of the prolific carbonate sequences of the T Group in another onshore field in GCC. The formation is characterized by moderate matrix permeability (typically 0.5–10 md), interbedded limestone and dolomitic intervals, and localized heterogeneities that can result in uneven acid placement and limited wormhole penetration when stimulated using conventional acid systems. The reservoir also exhibits tight vertical layering and occasional natural fractures, and a Bottom Hole Temperature of 280 DegF. The well's production history, made Well GP-01 a strong candidate for pilot deployment of the Single Phase Retarded Acid (SPRA) system.

Prior to the SPRA application, the well had undergone a stimulation treatment using conventional 28% hydrochloric acid (HCl), without reaching the production goals, but providing the basis for the comparison and assessment of the eventual SPRA treatment. This initial conventional acid treatment was conducted using 100 gal/ft dosage and no preflush was considered in the sequence, due to the very low injectivity, resulting in a modest increase of injectivity and moderate gas production improvement. The lack of pre-flush may have limited acid effectiveness due to poor near-wellbore conditioning, contributing to shallow wormhole penetration. Nonetheless, the HCl treatment likely removed some near-wellbore damage and initiated basic flow paths for future enhancement.

The subsequent SPRA treatment was designed with the A06 system and following confirmation of laboratory corrosion/retardation tests. The treatment followed below sequence:

1. **Pre-flush:** conventional HCl system was implemented to gain low injection rate as low as 0.2 bpm
2. **Main SPRA Acid Stage:** A06 acid system pumped at a controlled ramp-up rate to initiate slow reactivity and prevent premature acid spending at the wellbore.
3. **Overflush:** Fresh water post-flush to push the reacted acid deeper and displace any residual surfactant/acid mixture from the near-wellbore region.

During the SPRA treatment, pumping and rate development were closely monitored: initially, injection was maintained at 0.2 bpm, mirroring the maximum rate previously achieved with conventional acid systems. As the A06 acid was introduced at a 130 gal/ft dosage, injection rates climbed progressively, reaching 1.0 bpm within minutes and then stabilizing near 2.0 bpm as the wormhole network expanded. Following the main acid stage, fresh water was injected, resulting in sustained injectivity at 2.2 bpm and confirming the improvement in permeability and the stable conductivity of the newly developed wormholes.

As shown in Figure 10, the post-SPRA production response was immediate and substantial. The gas production rate increased to 2X at essentially the same WHFP, nearly double the rate achieved after the HCl treatment. This increase was sustained across multiple test chokes and pressure drawdowns, further validating the efficiency of wormhole development and flow enhancement achieved with the new system.

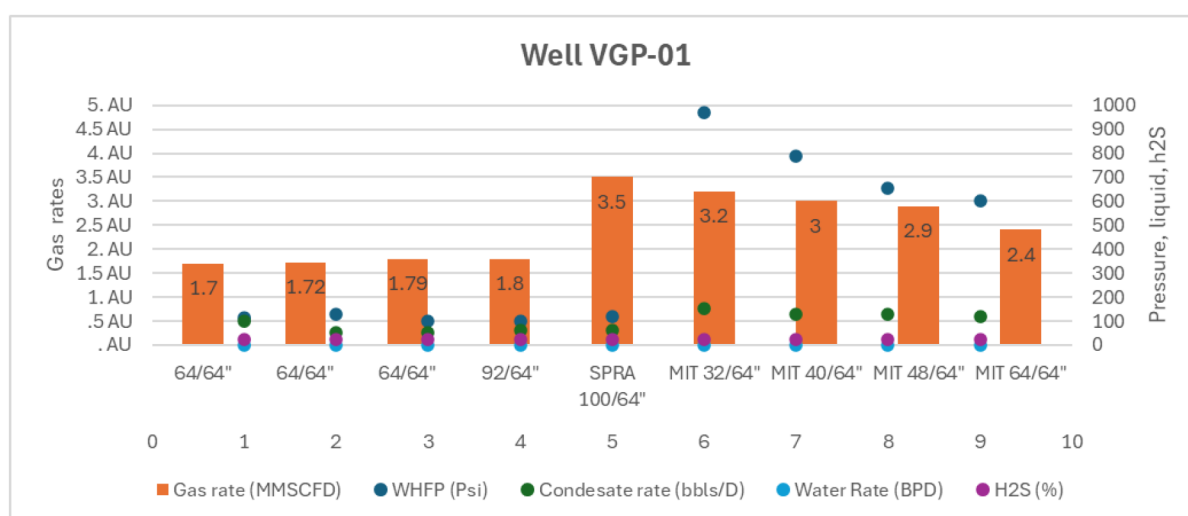


Figure 10—Well VGP-01 - Production Performance (A. U. = Arbitrary Units)

Figure 10 also shows the shift in gas productivity following SPRA injection at different choke sizes; as well the trend across subsequent Multi-Rate Injection Tests (MITs) confirms the stability of performance improvements over time.

A review of the operational experience with the SPRA treatment in the GP-01 well, highlights several important learnings that can guide future applications and optimization strategies for the same reservoir.

While SPRA was employed at a higher dosage of 130 gal/ft compared to the earlier treatment, the pronounced gains in production strongly indicate potential for further fine-tuning of treatment volumes in subsequent operations to maximize cost-effectiveness without diminishing outcomes. The inclusion of both pre-flush and post-flush stages in the treatment sequence proved to be instrumental in enhancing acid placement, injectivity, and overall stimulation efficiency, as these steps enabled more thorough conditioning and cleanup of the near-wellbore environment. Notably, there appears to be a synergistic benefit from the initial HCl stimulation, which may have reduced the formation's skin and facilitated deeper acid propagation during the SPRA phase, ultimately contributing to the marked improvements in gas flow and sustained productivity seen post-treatment. These insights collectively underscore the value of integrated and carefully sequenced acid stimulation protocols when aiming for optimal reservoir performance.

The economic evaluation conducted for the treatments compared the outcomes of the initial 28% HCl application to those achieved following the 15% HCl SPRA stimulation. The SPRA approach delivered a 1.5-FOI with respect to the regular system.

Financial projections for the vertical well establishes a 35% Increment in the revenue generated from enhanced production during the first year. In terms of direct costs, a detailed comparison was made between the SPRA acid blend—delivered with the A06 system—and the conventional 28% HCl formulation, showing less differential, as compared to 15% HCl systems. As a result, the projected NPV increase over the Business as Usual scenario remained very close 35% as well.

These results underline the economic and operational advantages of SPRA over traditional acid treatments, demonstrating its potential for both immediate and sustained production gains in gas reservoirs.

Results Analysis

From an operational standpoint, SPRA offers several key advantages that contribute to its superior effectiveness. The formulation enables deeper acid penetration into the reservoir, allowing the reactive fluids to reach and stimulate a greater volume of the formation. This deeper reach is coupled with enhanced wormhole propagation, resulting in more extensive and efficient flow paths within the carbonate matrix.

One of the most significant advantages observed with SPRA treatments is their marked superiority in enhancing both productivity and injectivity indices when compared to conventional acid systems. Across a range of well types, including oil producers, gas producers, and water injectors—SPRA consistently delivered higher performance. In particular, oil-producing wells that underwent SPRA stimulation often experienced significant increase in oil output, in some cases doubling the production rates previously achieved with traditional treatments. Similarly, for water injection wells, the improvements were even more pronounced; some cases reported up to a fourteen-fold increase in injectivity index, underscoring the system's capacity to significantly augment fluid intake and overall well throughput.

The economic impact of adopting SPRA methods is equally compelling. Field case studies have demonstrated that the incremental revenues generated from enhanced production are substantial, often surpassing the gains realized through conventional acidizing approaches. Although SPRA treatments may involve higher upfront costs due to more advanced formulations and specialized application protocols, these investments are typically offset by the impressive financial returns observed post-treatment. The return on investment (ROI) and net present value (NPV) calculations consistently favor SPRA, with strong evidence of increased profitability and long-term value creation. These economic outcomes, coupled with the operational and technical benefits, make SPRA a highly attractive option for operators seeking to maximize both immediate and sustained performance gains in carbonate reservoirs.

Conclusions and Future Directions

SPRA has demonstrated itself as a highly viable alternative, particularly in moderate-permeability carbonate reservoirs where both technical advantages and economic gains are pronounced. Its ability to outperform

traditional acid treatments in terms of productivity and injectivity indices underscores its role as an attractive solution for reservoir stimulation. Operators benefit not only from improved immediate results but also from measurable long-term production gains, positioning SPRA as a cornerstone technology in the evolving landscape of reservoir enhancement.

A key factor behind the success of SPRA treatments is the incorporation of data-driven design, notably the use of pore volume to breakthrough (PVbt) laboratory screening. This approach allows engineers to tailor acid systems precisely to the unique conditions of each reservoir, ensuring optimal acid placement, deeper penetration, and maximized stimulation efficiency. Such customization leads to more effective outcomes and helps mitigate the risk of suboptimal treatments, further improving return on investment.

Looking forward, there are several promising avenues for future work. Expanding SPRA applications to even tighter formations could unlock additional value, with the potential for breakthroughs in challenging reservoir environments. Long-term performance monitoring will be essential to fully understand the sustained benefits and to validate the durability of productivity improvements achieved with SPRA. In tandem, continued efforts to optimize treatment parameters—balancing cost-effectiveness with operational efficiency—will ensure that SPRA delivers consistently high returns across a broader range of reservoir conditions. Together, these recommendations lay a strategic foundation for further advancements and broader adoption of SPRA technology.

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